



Oculus primed

The Oculus Rift, hugely criticized for its less-than-perfect resolution takes another step for the consumers

Some chroming required

Google Chrome may be raising efficiency... But at what cost? Find out here: <http://dgit.in/180UjST>

which is actually smaller than the pixel pitch on most other camera phones. However, Nokia offsets this by a very intelligent algorithm that combines the data from four adjacent pixels into one. This allows the Lumia 1020 to pretty much nail everything on our resolution chart. The horizontal and vertical lines on our chart rendered beautifully on both the 38 megapixel and even the 5 megapixel files. The edge sharpness was pretty good, too as the text we placed in the corners was clearly readable. Frankly put, performance like this coming from a cellphone camera, one that we normally only see in a point and shoot camera, is impressive.

Speaking of point and shoot cameras, the Xperia Z1 was

not to be left behind. With the very impressive 16 megapixel 1/2.3-inch sensor paired with a Sony G lens, the Xperia Z1 managed to dish out some very impressive performance as well. Edge sharpness was surprisingly good, but what we liked the most was that it was able to do justice to the colours on our test. The Lumia 920 and the 925 did equally well, which was a little surprising seeing how the Lumia 925 has a sixth lens element that is supposed to improve the image quality on the camera. The iPhone 5 did quite well as far as the resolution goes, but had a little problem keeping the distortion low.

The LG G2, despite having the relatively smaller sensor when compared to the Lumia

1020 and the Sony Xperia Z1, did exceptionally well. Even though it had slightly lower



Sony Xperia Z1

resolving power than the other two, it still was on the higher side. If you chose the G2 over all others in this shootout, you'd still be getting great images. The Samsung Galaxy Zoom, is the most novel of the lot, and lends the feeling of shooting with an actual point and shoot, but doesn't do as well as a point and shoot would. The images out of the Galaxy Zoom visibly lacked edge sharpness, which is a little surprising seeing how the optical assembly on this big guy is the same one would find on a point and shoot.

The HTC One dual SIM comes with the "Ultrapixel" sensor, which is the same size as a regular camera sensor (1/3 inch) but with only 4 megapixels of resolution. HTC says what they don't deliver in megapixels, they claim to deliver in image quality, but unfortunately, our resolution chart begs to differ. The edge sharpness on the HTC One was disappointing, as was the colour reproduction. It did, however, manage to make the most out of the 4 megapixel sensor in terms of resolution reproduction though.

If a winner was to be picked, it would be the Lumia 1020 for sure, but the Xperia Z1 isn't too far behind. The Lumia 920 and 925 can hold their own with their 8 megapixel sensors, as can the LG G2 and the Samsung Galaxy Zoom with their 13 and 16 megapixel slabs of silicon respectively. The iPhone 5 is good too, but it showed some distortion while the HTC had trouble with the colours.

The real world dominator

Now this is where things got really interesting. While the image quality of the Nokia Lumia 1020 was top-notch, it seriously faltered in some

HOW WE TESTED

The goal of our test this time was to determine which phone camera would give the best image quality regardless of the shooting conditions. As a camera that's going to go everywhere with you, we also wanted to make sure the video features on these shooters was up to the task of documenting our all your precious moments.

We divided our tests into three parts, two for stills and one for video. One set of tests involved shooting a simulated test target in controlled light to measure color depth/accuracy, white balance accuracy and lens sharpness in the corners and center. A controlled test target also helps determine the ISO performance of a camera. We also devised our test target to measure dynamic range by including pure black, pure white and neutral grey areas which would effectively measure the sensor's ability to capture various levels of illumination. All test target shots were taken in Program mode as most of these cameras do not have an Aperture Priority mode. At the center of the test target is a resolution chart that helps us measure horizontal and vertical resolution.



This fine spreadout was our palette for the test.

The second phase of testing involved getting outside the studio and shooting with each of the cameras a "real world" scenario. We included several kinds of still life, lots of landscape, some macro along with the usual people shots. For our low light testing, we just walked around with all the phones late at night shooting the cityscape. We also shot coloured targets in a dimly lit room to see how good the colour accuracy would be in low light/high ISO conditions, along

with checking for sharpness across the board.

Video performance was tested outdoors during daytime and night. While the outdoor filming allowed us to gauge the dynamic range and audio prowess of each of the cameras, the night time test allowed us to have a true measure of how well the sensor can curtail and snub noise along with how much detail it could retain in highlight and shadow areas. All shooting was done using the default camera app.